FARAH ADRIANA BINTI MOHD SAMSUL

Seremban, Negeri Sembilan | +6013-674 5891 | farah.adriana11@gmail.com
LinkedIn: https://www.linkedin.com/in/farahadrianams/



EDUCATION BACKGROUND

2024 - Current **Universiti Teknologi Malaysia**
Master in Philosophy in Electrical Engineering

2019 - 2023 **Universiti Teknologi Malaysia**
Bachelor of Engineering (Electrical-Electronics)

WORK EXPERIENCE

Intelligent Circuit Engineering (ICE) **June 2026 - August 2026**
IC Design Trainee

- Designed, integrated, and verified an AMBA AHB-Lite peripheral subsystem with memory-mapped SPI registers using industry-standard simulation flows.
- Developed an AHB Bus Functional Model (BFM) for protocol-compliant read/write verification.
- Built synthesizable RTL and testbenches for AHB-Lite pipelined transfers and peripheral communication.
- Debugged functional and timing issues using simulation waveforms.

Universiti Teknologi Malaysia **October 2024 - August 2026**
Graduate Research Assistant

- Designed and optimized a fiber optic daylighting system to improve indoor natural light distribution and reduce reliance on artificial lighting.
- Conducted ray-tracing simulations using Zemax to model multi-stage concentrator performance, evaluating illuminance distribution, and optimizing system efficiency.
- Iteratively refined designs based on simulation results, achieving measurable improvements in performance and feasibility.
- Processed and analyzed large simulation datasets to identify trends and support data-driven design decisions.

Projek Lebuhraya Usahasama Berhad **February 2024 - October 2024**
Graduate Trainee (Toll Assurance & Recovery)

- Designed and maintained a Power BI dashboard to deliver detailed insights on monthly recovery data to the management.
- Provided administrative support for the Systematic Toll Outstanding Recovery Management (STORM) platform, addressing user inquiries and resolving account-related issues.
- Monitored and recorded daily payment transactions, ensuring accuracy and timely system updates.
- Conducted quarterly compliance audits to verify that plaza operations are aligned with documented working instructions, enhancing compliance and minimizing operational risks.

Medivest Sdn Bhd - Hospital Tuanku Ja'afar **July 2022 - October 2022**
Engineering Trainee (Electrical)

- Managed energy consumption data and monitored retrofit projects to optimize resources.
- Monitored the retrofit project to ensure it was completed on time, within budget, and met all stakeholders' requirements.
- Prepared and presented comprehensive maintenance reports to facility management, highlighting key performance indicators and areas for improvement.

PROFESSIONAL DEVELOPMENT

Advanced Semiconductor Academy of Malaysia (ASEM) - (April 2026)
National Semiconductor Excellence Program

AI for MY Future Upskilling Program - (2025)

Train & Place Upskilling Program - (2023)
Embedded Systems & IoT

SKILLS & INTERESTS

- **Engineering Software:** MATLAB, AutoCAD, Zemax, Synopsys IC Compiler II
- **Programming Languages:** Verilog, C
- **Data Analysis & Visualization:** Power BI, Tableau

LANGUAGES

- **Malay** (Speaking/Writing) Native
- **English** (Speaking/Writing) Proficient

REFERENCE

Student Affairs
Advanced Semiconductor Academy of Malaysia (ASEM)
+603-8320 9499